Rational Functions: Domain and Range

Questions 1–19

Question 1. The function f is defined by $f(x) = \dfrac{2x+3}{x-1}$.

a) State the domain of f.
b) State the equations of any asymptotes of the curve $y = f(x)$.
c) Find the range of f.
d) Sketch $y = f(x)$.

Question 2. The function f is defined by $f(x) = \dfrac{3x-2}{4-3x}$.

a) State the domain of f.
b) State the equations of any asymptotes of the curve $y = f(x)$.
c) Find the range of f.
d) Sketch $y = f(x)$.

Question 3. The function f is defined by $f(x) = \dfrac{x^2 - 1}{x^2 + 4}$.

a) State the domain of f.
b) State the equations of any asymptotes of the curve $y = f(x)$.
c) Without using Calculus, find the range of f.
d) Sketch $y = f(x)$.

Question 4. The function f is defined by $f(x) = \dfrac{3x^2 - x - 2}{x^2 - x - 2}$.

a) Find the equations of all asymptotes of $y = f(x)$.
b) Without using Calculus, find the range of f.
c) Find the coordinates of the turning points.
d) Sketch the curve.

Question 5. The function f is defined by $f(x) = \dfrac{x^2 - 1}{(x+2)^2}$.

a) State the domain of f.
b) Find the equations of all asymptotes of $y = f(x)$.
c) Without using Calculus, determine whether the function has a maximum or minimum value.
d) Find the range of f.
e) Sketch $y = f(x)$.

Question 6. Given that $f(x) = \dfrac{ax^2 + 6}{x^2 + 6x + 5}$, where a is a positive constant:

a) State the domain of f.
b) State the equations of any asymptotes of the curve $y = f(x)$ in terms of a.

Given that the horizontal asymptote is $y = 3$:

a) Find the y-coordinates of the turning points.
b) Find the range of f.

Question 7. The function f is defined by $f(x) = \dfrac{2x^2+3}{x^2+1}$.

a) State the domain of f.
b) Without using Calculus, find the range of f.
c) Giving your answer in set notation, solve $f(x) \geq \dfrac{5}{2}$.

Question 8. The function f is defined by $f(x) = \dfrac{x^2-6x+16}{x^2-4}$.

a) State the domain of f.
b) Without using Calculus, find the range of f.
c) Sketch the curve $y = f(x)$.
d) Solve the inequality $-4 \leq f(x) < 1$.

Question 9. For constant a, the function f is defined by $f(x) = \dfrac{x^2 + ax + 4}{x^2 + 1}$.

a) Show that if $f(x) \geq 0$, then $a^2 \leq 16$.
b) For $a = 4$, without using Calculus, find the range of f.
c) For $a = 4$, solve $f(x) \leq 2$.

Question 10. Two curves C_1 and C_2 are defined by

$$C_1 : y = \frac{6}{x^2 - 5x + 6}, \qquad C_2 : y^2 = \frac{6}{x^2 - 5x + 6}.$$

a) Sketch C_1.
b) Determine the exact coordinates, if they exist, at which C_1 and C_2 intersect.
c) Hence solve $\dfrac{6}{x^2 - 5x + 6} > \sqrt{\dfrac{6}{x^2 - 5x + 6}}$.

Question 11. The function f is defined by $f(x) = \dfrac{x^2+1}{x}$.

a) State the equations of any asymptotes of the curve $y = f(x)$.
b) Without using Calculus, find the range of f.
c) Solve $f(x) > x$.

Question 12. The function f is defined by $f(x) = \dfrac{x^2-3}{x+2}$.

a) State the equations of any asymptotes of the curve $y = f(x)$.
b) Without using Calculus, find the range of f.
c) Solve $f(x) \leq x - 2$.

Question 13. The function f is defined by $f(x) = \dfrac{3x^2 - 2x + 19}{x - 2}$.

a) State the domain of f.
b) State the equations of any asymptotes of the curve $y = f(x)$.
c) Without using Calculus, find the range of f.

Question 14. The function f is defined by $f(x) = \dfrac{2x^2 + 5x - 3}{x + 1}$.

a) State the domain of f.
b) State the equations of any asymptotes of the curve $y = f(x)$.
c) Without using Calculus, find the range of f.

Question 15. Given that $f(x) = \dfrac{x^2 + ax + 1}{x + 4}$, where a is a constant:

a) State the domain of f.
b) Find the value of a such that the graph has an oblique asymptote of $y = x$.
c) For the value of a found in part (b), without using Calculus, find the range of f.

Question 16. Given that $f(x) = \dfrac{x^2 + ax + 4}{x - 3}$, where a is a constant:

a) Find the equation of the oblique asymptote in terms of a.
b) Find the value(s) of a such that the oblique asymptote passes through the point $(0, -1)$.

For the value of a found in part (b):

a) Find the domain of f.
b) Without using Calculus, find the range of f.
c) Sketch $y = f(x)$.

Question 17. The function f is defined by $f(x) = \dfrac{2x^2 + x - 6}{x - 2}$.

a) State the equations of any asymptotes of the curve $y = f(x)$.

b) Without using Calculus, find the range of f.

c) Solve $f(x) > 2x + 6$.

Question 18. The function f is defined by $f(x) = \dfrac{x^2 + ax - 1}{x + 1}$, where a is a constant.

a) State the vertical asymptote and find the equation of the oblique asymptote in terms of a.

b) Find the value of a such that $f(2) = \dfrac{1}{3}$.

c) For the value of a found in part (b), without using Calculus, find the range of f.

d) For the value of a found in part (b), solve $f(x) \geq x + 1$.

Question 19. Two curves C_1 and C_2 are defined by $y = f(x)$ and $y = g(x)$ respectively, where

$$f(x) = \frac{x^2 - 2x + 5}{x - 1}, \qquad g(x) = \big(f(x)\big)^2.$$

a) State the equations of any asymptotes of the curve $y = f(x)$.
b) Without using Calculus, find the range of f.
c) Sketch the curve C_1, including any turning points on your diagram.
d) Determine the exact values, if they exist, at which C_1 and C_2 intersect.
e) Hence solve $\dfrac{x^2 - 2x + 5}{x - 1} > \sqrt{\dfrac{x^2 - 2x + 5}{x - 1}}$.